



Michal Stroff

GYBU, 7.B



Space, Time & Matter

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Is it a symbiosis?

symbiosis *noun*

 /,sɪmˈbaɪəʊsɪs/

 /,sɪmˈbaɪəʊsɪs/

[uncountable, countable]

(plural **symbioses**  /,sɪmˈbaɪəʊsɪz/  /,sɪmˈbaɪəʊsɪz/)

- 1 ★ (*biology*) the relationship between two different living creatures that live close together and depend on each other in particular ways, each getting particular benefits from the other

- *The birds live in symbiosis with the cattle, picking insects from their skin to eat.*

TOPICS **Biology** **C2**

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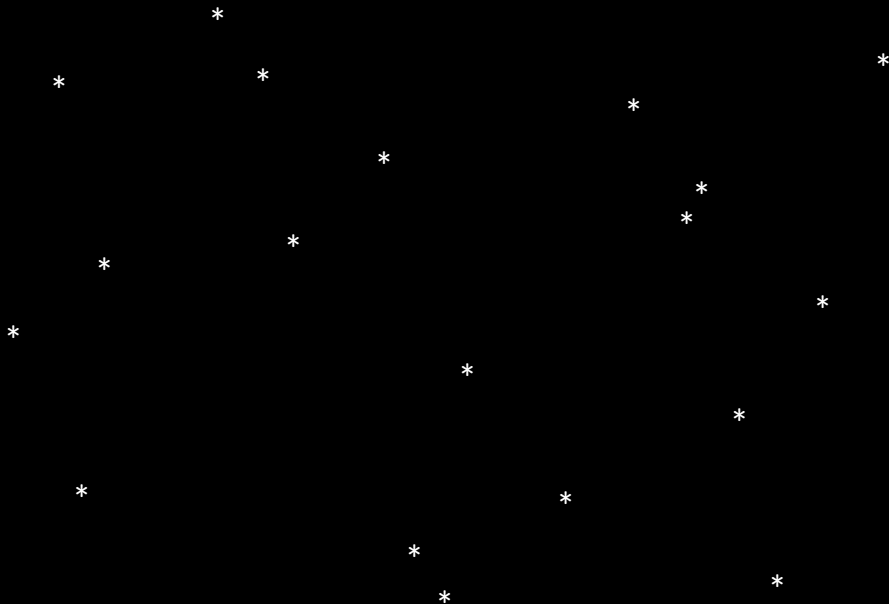
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- ▶ *symbiosis between matter, space and time* – generalization of all the other symbioses :D
- ▶ *physics* – generalized biology???

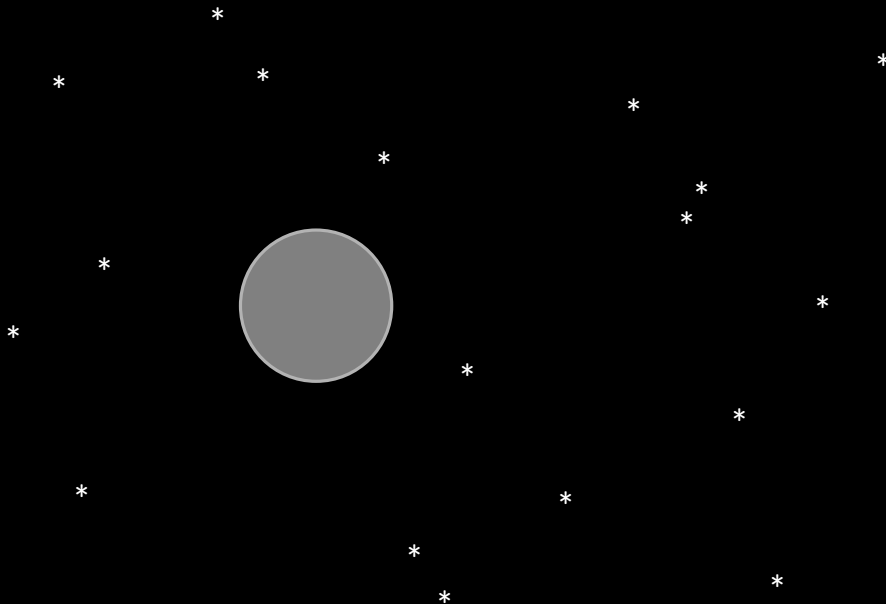
But...

Where is the *coexistence* of space,
time and matter?

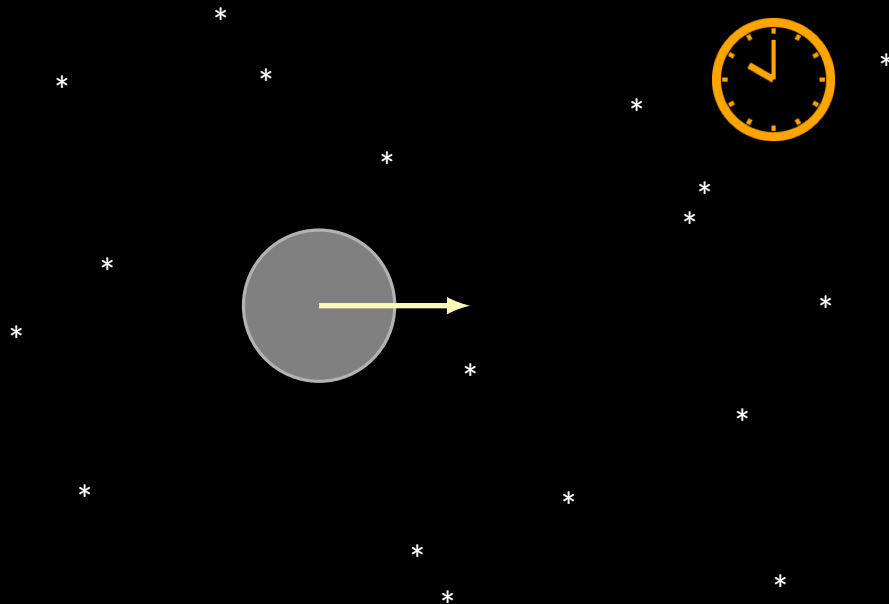
Gedankenexperiment

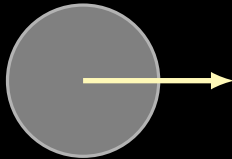


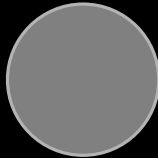
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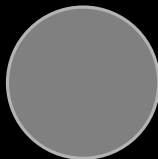
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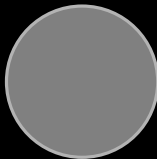


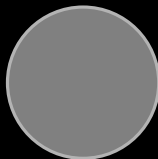


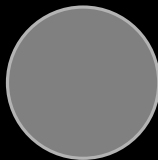


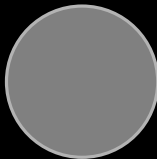
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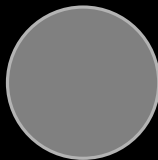








Gedankenexperiment



Gedankenexperiment

NO TIME

NO TIME

NO SPACE

Mach's principle

- ▶ criticism of Newton's *absolute space and time*
- ▶ Acceleration and velocity of a body only has a good sense if it is defined and measured with respect to other bodies.

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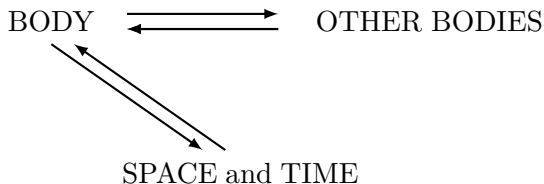
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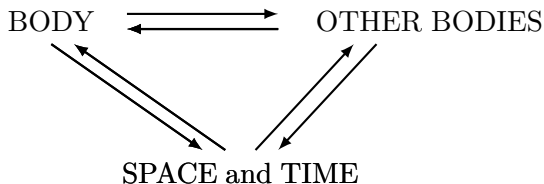
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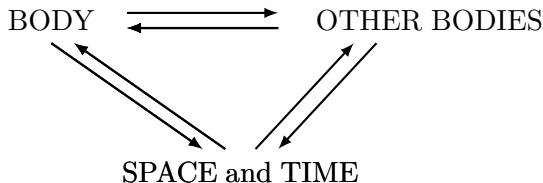
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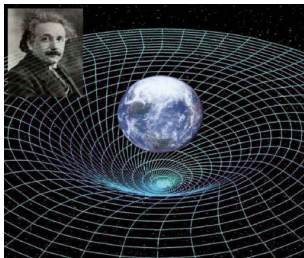


MATTER $\longleftrightarrow$ SPACE and TIME

Einstein's general relativity

SPACE and TIME $\longleftrightarrow$ MATTER

$$G_{\mu\nu} = \kappa T_{\mu\nu}$$



Matter tells space-time how to curve, and space-time tells matter how to move.

— John Wheeler

Space and time cannot exist without matter and vice versa. They coexist in a *symbiosis*.